

Software Requirements Specification (SRS)

Grocery Ordering & Inventory Management System

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for the Grocery Ordering & Inventory Management System, consisting of an Admin Web Portal, Backend API, and Customer Mobile App.

1.2 Scope

The system enables the grocery store to manage products, inventory, orders, purchase orders, customers, and payments. Customers can browse items, place orders, and track delivery through the mobile app.

2. Overall Description

2.1 Product Perspective

The system consists of:

- Admin Portal (ASP.NET Core MVC)
- Backend API (ASP.NET Core Web API)
- Mobile App (Flutter)
- Database (SQL Server)

2.2 User Classes

- Super Admin
- Staff/Admin User
- Customer
- Delivery Personnel

3. System Features

3.1 Admin Portal

- Product Management
- Inventory Management
- Purchase Orders
- Order Management
- Customer Management
- Reports & Analytics

3.2 Mobile App

- User Registration/Login
- Product Browsing
- Cart & Checkout
- Payment (COD/Online)
- Order Tracking

4. Functional Requirements

4.1 Product Management

- FR1: Admin can add/edit/delete products.
- FR2: Admin can assign categories and subcategories.
- FR3: Admin can upload product images.

4.2 Inventory Management

- FR4: System updates stock automatically on order placement.
- FR5: System alerts for low stock.
- FR6: Admin can manually adjust stock.

4.3 Order Management

- FR7: Admin can accept/reject orders.
- FR8: Admin can track payment status.
- FR9: System updates order status through each stage.

5. Non-Functional Requirements

- NFR1: System should support up to 10,000 products.
- NFR2: API should respond within 2 seconds.
- NFR3: Secure authentication with JWT.
- NFR4: Data should be backed up daily.

6. System Architecture

- REST API
- MVC-based Admin Portal

- Flutter-based Mobile App
- SQL Server Database

7. Constraints

- Internet required for mobile app use.
- Hosting server must support .NET Core applications.

8. Assumptions

- Admin users are trained.
- Customers have smartphones with internet.

9. Acceptance Criteria

- System functions correctly across modules.
- API integrates seamlessly with mobile app.
- Admin portal is accessible and bug-free.